Forge Conductor - Goals, Integrations and Decision Plan

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GOALS - the durable focus for this program

Status: authoritative for the research/implementation program. Read this before any plan or code.

Date: 2026-11-05 - **Owner:** Riley - **Program:** DeerFlow Forge / Forge Conductor.

Mission

Build a **hosted, auditable design-production agent** that can take a brief, use the installed Craft apps through their native command/MCP surfaces, verify its own output with hard evidence, remember what worked, and ship only what a human has approved.

It is not "a chatbot that talks about design." It is a production operator with a ledger.

Product outcome (what exists when this succeeds)

A user can submit a print/vector/web brief to **Forge Conductor**. It:

1. selects the right app and exact command ids from the pinned catalogue;
2. runs on a copy, records every action and artifact;
3. verifies dimensions, DPI, colour, gates and a blind visual review;
4. refuses or asks a human when evidence is missing or the taste call is unresolved;
5. stores a case/skill candidate from the run;
6. promotes a procedure only after repeated verified success + regression + human approval;
7. never autonomously publishes to the storefront.

Non-negotiable goals

ID	Goal	Success test
G1	Execute the installed Craft apps reliably	one VectorCraft trace and one PhotoCraft retouch run from the pipeline path, with expected artifacts
G2	Use native commands/MCP, not blind clicking	every action is a catalogued command id or a justified browser fallback
G3	Verify with evidence	artifact hash + dimensions/DPI + gates + independent render + blind critic + human where applicable
G4	Remember and improve truthfully	L1/L2 ledger/retrieval/skills; A/B shows gain or is disabled
G5	Host on the existing VPS/model	loopback services, 2 vCPU budget, DeepSeek key by env name, no new public port
G6	Keep humans on taste/publish	approval is required for print/publish and skill activation
G7	Be honest about limits	VERIFIED/READ/INFERRED/UNVERIFIED labels; no "better than human" claim without a study

Non-goals / explicit boundaries

- **No claim** that the agent is better than a human designer; target is a reliable production operator under human art direction.
- **No weight training, fine-tuning or RL** in Phases 0-2; no GPU/open weights on this VPS.
- **No autonomous publish** or storefront mutation.
- **No fake 1:1 Adobe compatibility**; the Craft apps are independent implementations.

- **No all-app claim** until FilmCraft/DesignCraft/EffectCraft are installed and catalogued.
- **No GUI-first agent** while there is no display/GPU and the control bridge is broken.
- **No unverified success** - exit 0, `ok:true` or a screenshot alone is not proof.

Scope (current)

```
| In scope | Out of scope (now) |
|---|---|
| VectorCraft 0.21.1 | FilmCraft (not installed) |
| PhotoCraft 0.20.2 | DesignCraft / EffectCraft (not installed) |
| Native CLI + MCP | Pixel-only desktop automation |
| Print/vector/raster verification | CMYK press proof without vendor proof |
| L1/L2 memory and skills | L3/L4 training/RL |
| Human-approved publish | Autonomous publish |
```

Success metrics

- First-attempt verified-pass rate on held-out goldens.
- Attempts per accepted artifact.
- Human rework minutes.
- Cost/tokens per accepted artifact.
- Unsafe publishes = **0**.
- Source mutations = **0**.
- Promotion uses a pre-declared evaluation horizon and Wilson lower bound above baseline; a 3/3 sample is a minimum, not proof (3/3 LB ? 0.4385).
- Every accepted artifact has a complete evidence chain.

Invariants

1. Work on copies; never overwrite the source/master.
2. Exact command id from the pinned catalogue; unknown id -> stop.
3. A failed gate is terminal for shipping; never edit a report to pass.
4. A screenshot or exit 0 can only downgrade confidence, never certify.
5. Memory is data, not instructions; no tool-scope expansion from retrieval.
6. A skill is promoted only by evidence + regression + human approval; rollback exists.
7. Secrets never enter prompts, logs, ledger, repo or screenshots.
8. The `pod` DB/pipeline remains deterministic; Forge may call it, not absorb it.

Current truth (2026-11-05)

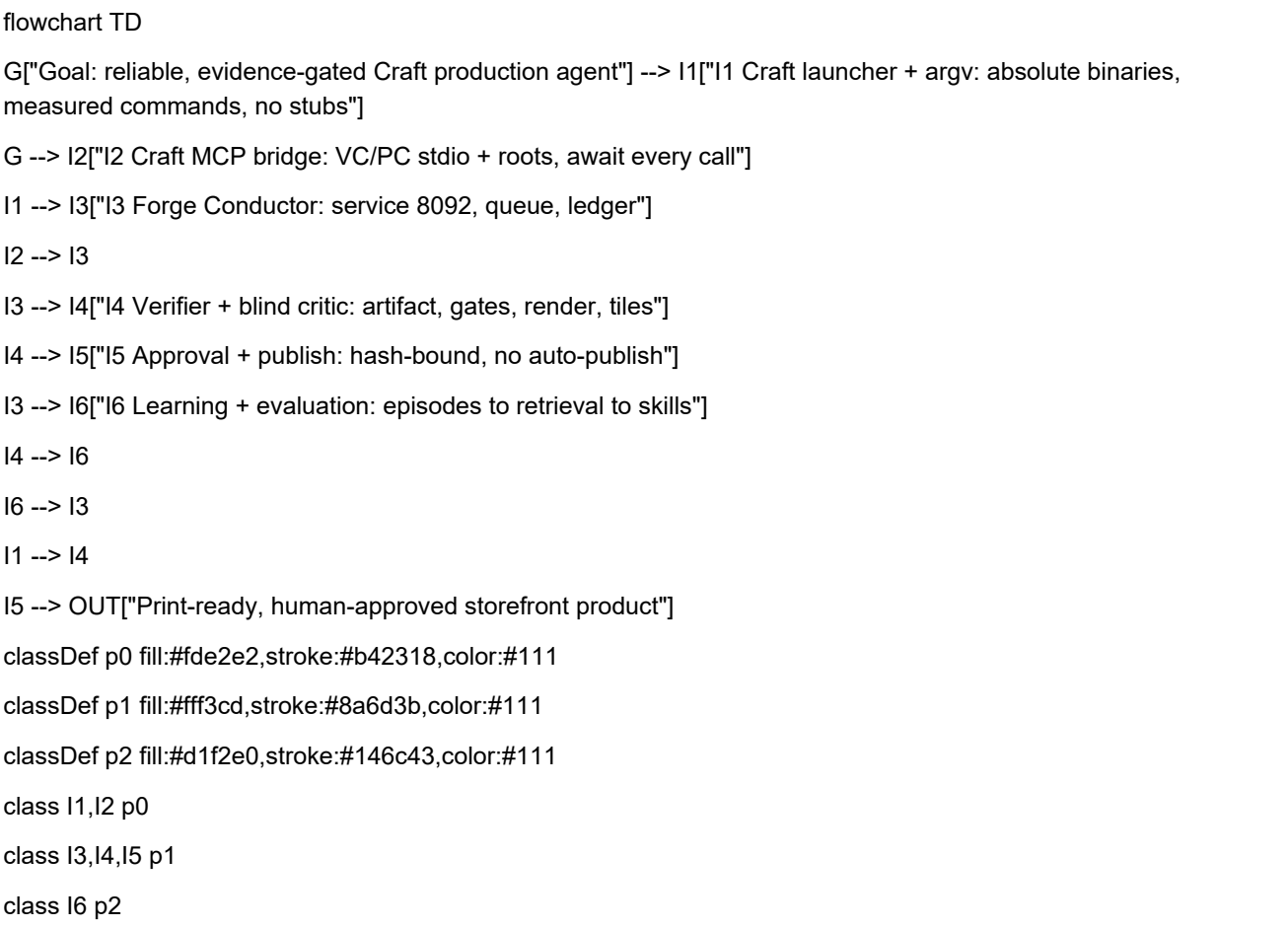
- VPS has 2 vCPU/7.9 GB, no GPU/display; DeerFlow 2.2 runs `deepseek-flash` (V4.1 Flash), key by env name only; key validity unverified.
- VectorCraft/PhotoCraft installed; Film/Design/Effect absent.
- `/opt/pod-pipeline/bin/*-cli` are stubs exiting 127; `routes.py` argv is invented; Craft MCP is not registered in DeerFlow.

- `pod` DB empty; no learning/eval tables; console is a viewer.

- 39 skills, 400 answers, six examples, eight diagrams - research/design, not deployed.

Decision log

Date	Decision
2026-11-05	Use DeerFlow 2.2 + DeepSeek as substrate; build a third `forge-*` plane, not a new harness
2026-11-05	Phase 0 fixes execution (shims/argv/MCP) before any agent feature
2026-11-05	L1/L2 only; no fine-tune/RL; no autonomous publish
2026-11-05	Target = reliable production operator under human direction, not "better than human"
2026-11-05	Promotion calibration: 3/3 is not sufficient; declare horizon + Wilson baseline; demotion/regression suite must be tested (task-11)


flowchart TD
G["Goal: reliable, evidence-gated Craft production agent"] --> I1["I1 Craft launcher + argv: absolute binaries, measured commands, no stubs"]
G --> I2["I2 Craft MCP bridge: VC/PC stdio + roots, await every call"]
I1 --> I3["I3 Forge Conductor: service 8092, queue, ledger"]
I2 --> I3
I3 --> I4["I4 Verifier + blind critic: artifact, gates, render, tiles"]
I4 --> I5["I5 Approval + publish: hash-bound, no auto-publish"]
I3 --> I6["I6 Learning + evaluation: episodes to retrieval to skills"]
I4 --> I6
I6 --> I3
I1 --> I4
I5 --> OUT["Print-ready, human-approved storefront product"]
classDef p0 fill:#fde2e2,stroke:#b42318,color:#111
classDef p1 fill:#fff3cd,stroke:#8a6d3b,color:#111
classDef p2 fill:#d1f2e0,stroke:#146c43,color:#111
class I1,I2 p0
class I3,I4,I5 p1
class I6 p2

09 - Goal-to-Code Gap Audit and Decision-Complete Plan

Date: 2026-11-05 - **Status:** PLAN, not implementation. No production file/DB/service changed. **Inputs:** `GOALS.md`, first `REPORT.md`, `wave2/08-200-QUESTIONS.md`, `wave2/01`-`07`, execution-wave plan, live VPS probes, local source. **Lens:** goal -> research -> code -> product -> operations.

1. Outcome

A hosted **Forge Conductor** on the existing DeerFlow 2.2/DeepSeek runtime that can execute a brief on VectorCraft/PhotoCraft through exact CLI/MCP commands, verify the artifact and gates, record an evidence-grade episode, propose only verified lessons, and require human approval to publish. The first release is a **production operator**, not an autonomous artist and not a new generic harness.

What users can do after Phase 1 that they cannot do now: submit a brief, watch a real trace/retouch run end-to-end, inspect the artifact hash + gates + blind critique, and reproduce the run from the ledger - with no stub, no invented argv and no fake success.

2. Current behavior (evidence, no assumptions)

#	Current behavior	Evidence	Status
C1	Pipeline cannot execute any Craft step	`/opt/pod-pipeline/bin/*-cli` 122-127 B shims exit 127; last run failed there	`VERIFIED`
C2	The argv contract is invented	`routes.py:73-80` (`trace`, `--input`, `--svg-out`) and `:88-96` (`--input` for PhotoCraft); real CLIs reject them	`VERIFIED`
C3	Only two Craft apps are installed	VectorCraft 0.21.1 / PhotoCraft 0.20.2; Film/Design/Effect absent host-wide	`VERIFIED`
C4	Craft MCP is not registered in DeerFlow	only `mcpServers.forge` exists; craft MCP handshakes work standalone	`VERIFIED`
C5	DeerFlow 2.2 + DeepSeek run on the VPS	backend 2.2.0; `deepseek-flash`/"V4.1 Flash", 1M ctx, 8k output; key by env name	`VERIFIED` config; key validity `UNVERIFIED`
C6	No learning/eval ledger exists	`pod` has 5 tables, no episode/action/skill/eval; DB empty	`VERIFIED`
C7	Console is a viewer	pure viewer at 127.0.0.1:8091; no job controls/auth	`VERIFIED`
C8	README's live E2E claim is not reproducible now	`README.md:191-194`; DB empty and stub failure; it may be historical	`HISTORICAL/UNREPRODUCIBLE`
C9	Gates can lie if read naively	`craft_trace:344` saves before gates; `:250` vacuous retype gate; `prep_print_art:356` `ok:True` with warning-only upscale	`READ` + prior probes
C10	MCP endpoint is host-sensitive	`pod-mcp` 0.0.0.0:8765, unauthenticated, pod in Docker group	`VERIFIED`
C11	39 skills/400 answers are instructions, not runtime	installed skills; pipeline has no LLM/loader	`VERIFIED`
C12	Foreign `AGENTS.md`	Grok App Builder contract (`/workspace`, port 8080) not this VPS	`READ`

2.1 Measured Phase 0 contracts (do not revert to the old ASSUMPTIONs)

Contract	Measured behavior
VectorCraft trace	`run --in F --cmd select.all --cmd imageTrace.make --params '{"params":{...}}' --cmd imageTrace.expand --export out.svg` -> exit 0, `{"result":{"bytes":334,"format":"svg"},"step":"export"}`; PDF export also works; **export does not create parent dirs** (`os error 2`)
PhotoCraft run	`run <FILE|--new> --cmd ID --params J --out F`; opens traced SVG with a benign flatten warning; `image.trim` 161x121; `image.imageSize` at 300 ppi; creates parent dirs

| VectorCraft MCP | `mcp --headless --automation-read-root ... --automation-write-root ...` |

| PhotoCraft MCP | `mcp` **without `--headless`**, same roots; dependent calls must be awaited (server can interleave) |

| Gateway/binary | gateway uid 0, glibc 2.36 >= binaries' max GLIBC_2.35; bind mount is viable |

| Security | `POD_MCP_HOST=0.0.0.0` in `pipeline.env` overrides code default `127.0.0.1`; rebind is one value. Key validation: `GET /user/balance` + `GET /models` with Bearer; 200 valid, 401 invalid, 402 no balance; **never** use a chat completion as a key test |

| Visual defect | global SSIM catches 7/15; tile-min catches 13/15 but flags 3/4 clean controls; region SSIM catches 11/15 with 0/4 false alarms - combine region + tile + crop, not one threshold |

3. Goal->code gap matrix

| Goal | Research says | Code/runtime reality | Gap | Severity | Owner |

|---|---|---|---|---|---|

| G1 execute Craft | real 918/965 registries, MCP works | stubs, wrong argv, MCP unregistered | execution path broken | **P0** | task-8 |

| G2 native tools | use CLI/MCP, not GUI | no gateway; GUI bridge broken | no typed tool surface | **P0** | task-9/13 |

| G3 verify | six-rung evidence ladder | gates exist but can be vacuous/pre-artifact; no critic | no verifier/critic integration | **P0** | task-10 |

| G4 learn | L1/L2 buildable, L3/L4 not | no ledger/eval/retrieval/skill tables | no learner | **P1** | task-11 |

| G5 host | VPS capacity, model exists | 2 vCPU/7.9 GB; key unverified; no GPU | capacity/key risk | **P1** | task-12 |

| G6 approval | human gate | no approval table/queue; publish config unset | no safe publish | **P0** | task-9/12 |

| G7 truth | labels/evidence | README claim drift; foreign AGENTS | documentation must be corrected | **P1** | Lead |

| All apps | only VC/PC installed | Film/Design/Effect absent | scope must exclude | **P1** | Lead |

| Better-than-human | no benchmark | no eval | claim must be dropped | **P0** | Lead |

4. Decisions (rejected alternatives)

| # | Decision | Rejected | Why |

|---|---|---|---|

| D1 | DeerFlow 2.2 is the runtime/front door | build a new harness; use Eko as core | already hosted, MIT, memory/subagents/checkpoints; Eko is dormant/JS/in-memory |

| D2 | Build a **third `forge-*` plane** | embed LLM in pipeline/console | preserves deterministic pipeline and replaceable agent |

| D3 | Phase 0 fixes execution first | build ledger/agent first | without real commands, every downstream test is fake |

| D4 | VC + PC only for MVP | all Craft apps | others absent; claiming otherwise is false |

| D5 | Native CLI/MCP only in MVP | GUI/computer-use first | no display/GPU; bridge panics; commands are deterministic |

| D6 | Postgres `forge` DB + queue | new broker; reuse `pod` DB | one DB, no new service, `pod` untouched |

| D7 | L1/L2 memory + skills; promotion calibrated (no 3/3 shortcut) | fine-tune/RL now | API model, no GPU, no corpus/eval; simulation shows 3/3 LB 0.4385, so horizon + baseline required; claim discipline |

| D8 | Human-approved publish | autonomous publish | irreversible external side effect; template ids missing |

| D9 | Blind critic + crop evidence | one global SSIM/screenshot | measured 4% defect changes SSIM only 0.99994->0.9787 |

| D10 | Use donor mechanisms, original adapters | copy/vendor whole repos | maintenance/licence/verifiability; no verbatim code |

5. File-by-file change (planned; no production edit in this task)

5.1 Phase 0 - execution repair

| Path | Change | Test |

|---|---|---|

| `blueprint/files/deploy/craft-bin/*-cli` | replace 127 shims with launchers to `/opt/craft/*/bin` (absolute); keep refusal if absent | `--version` for VC/PC; absent app -> named refusal |

| `blueprint/files/pod_pipeline/routes.py:73-80` | replace trace argv with `run --in {src} --commands {steps} --export {svg}` or dedicated wrapper | one real trace; artifact exists |

| `blueprint/files/pod_pipeline/routes.py:88-96` | replace PhotoCraft `--input` contract with `photocraft-cli run FILE ... --out` | one real retouch |

| DeerFlow `extensions_config.json` | register vectorcraft/photocraft MCP stdio with roots; no arbitrary host paths | `tools/list`, `command_batch` |

| `README.md:188-211` | relabel live-E2E as historical/unreproducible; state current blockers | doc read-back |

| `AGENTS.md` | replace foreign Grok contract with a DeerFlow Forge/VPS contract, or remove | no port-8080 claims |

5.2 Phase 1 - Forge skeleton

| Path | Change |

|---|---|

| `blueprint/files/forge/contracts.py` | TaskEnvelope, ArtSpec, Handoff, ToolCall, Artifact, Verdict (versioned) |

| `blueprint/files/forge/gateway.py` | absolute binary registry, allowlist/roots/idempotency/refusals |

| `blueprint/files/forge/verifier.py` | artifact hash/dimensions/DPI + gate-report parser + read-back |

| `blueprint/files/forge/jobs.py` | Postgres queue `FOR UPDATE SKIP LOCKED`, leases, attempts |

| `blueprint/files/sql/schema_forge.sql` | new `forge` DB tables: episode/event/handoff/tool_call/artifact/gate/critique/eval/skill/job/budget/role |

| `blueprint/files/forge/critic.py` | blind prompt/schema, crop grounding, cannot-assess path |

| `blueprint/files/forge/memory.py` | episode append, bounded retrieval, candidate lessons |

| `blueprint/files/forge/promotion.py` | Wilson bound, N>=3, regression, approve/rollback |

| `deploy/systemd/forge-conductor.service` | loopback 8092, `ProtectSystem=strict`, env file |

| `deploy/systemd/forge-worker.service` | queue worker, <=2 concurrency, one craft semaphore |

| `deploy/systemd/forge-ui.service` | optional 8093 operator approval UI (loopback) |

5.3 Phase 2 - learning and evaluation

| Path | Change |

|---|---|

| `blueprint/files/forge/eval.py` | golden runner, A/B arms, metrics, JSON report |

| `blueprint/files/forge/evaluation/` | 10-50 golden fixtures (rights-cleared) |

| `blueprint/files/skills/*` | the five research skills + operator; loaded by agent, not pipeline |

| DeerFlow memory config | keep DeerMem for facts; Forge ledger for episodes/skills; do not duplicate |

5.4 Never change

`pod` schema, orchestrator determinism, console viewer contract, Craft app binaries/catalogues,

publish without approval, any secret value in repo/logs.

6. Failure modes and required behavior

| Failure | Required behavior |

|---|---|

| Missing/stub binary | named `E_APP_UNAVAILABLE`; no fallback that pretends success |

| Wrong argv/unknown id | `E_COMMAND_UNKNOWN`/`E_PARAM_REJECTED`; stop |

| No document/selection | open/select first; `enabled:false` is state, not absence |

| PhotoCraft concurrent MCP | await each reply; never pipe dependent calls |

| Artifact exists but gate failed | `failed`; preserve artifact for diagnosis; no ship |

`prep_print_art.ok` true with warning	parse `warnings`; over-2.5x upscale is a refusal for print
Text gate with 0 boxes	`unverified`, not pass
Source mutated	fail episode; quarantine
Key missing/invalid	named refusal; no fake completion; owner-approved live test
Budget exceeded	stop before spend; `E_BUDGET`
Worker crash	lease expiry/requeue; idempotent tool key; no duplicate publish
Critic/gate disagree	`needs_human`; never critic override
Publish without approval	refuse; `E_PUBLISH_NOT_APPROVED`
App/catalogue drift	pin hash; skill expired until revalidated

7. Tests (discriminating, including error paths)

Test	Command / evidence	Pass
T0 versions	`vectorcraft-cli --version`, `photocraft-cli --version`	expected versions
T1 trace	real 1024? badge via `craft_trace`/run	SVG/PNG + report, `ok:true`, 10 gates
T2 retouch	PhotoCraft crop->filters->imageSize on copy	2400x3000, 118.11 px/cm
T3 MCP	initialize -> tools/list -> command_batch	33/26 tools; batch result
T4 refusals	unknown id/outside root/missing doc	named codes; no write
T5 duplicate	same idempotency key twice	one effect, same result
T6 crash/resume	kill worker mid-episode	lease requeue; no duplicate
T7 verifier	fixture with failed gate	`failed`, not ship
T8 critic	labelled defect set	precision/recall + false-pass reported
T9 promotion	calibrated horizon (e.g. n=8, k>=7, LB>p0) + regression + human	simulation shows null-promotion rate and retention; 3/3 rule rejected; no overclaim
T10 packaging	copy corpus/skills	links/JSON/validation clean
T11 docs	README claims	no unreproducible "live E2E"

8. Rollback

- **Phase 0:** restore original `deploy/craft-bin` shims, `routes.py`, DeerFlow MCP config from backup; no DB migration.
- **Phase 1:** stop/disable `forge-*` units; `DROP DATABASE forge`; artifacts remain content-addressed; `pod` untouched.
- **Phase 2:** feature flag `FORGE_MEMORY=off`; retrieval/skills disabled; prior skill versions retained.
- Every file change has a pre-change SHA-256; rollback is a pointer/file restore, not a rewrite.
- If a live key test was approved, record spend; rotate the key if exposed.

9. Out of scope / blockers

- FilmCraft/DesignCraft/EffectCraft not installed; no command claims.
- GUI/computer-use blocked by missing display and broken control bridge.
- No fine-tune/RL (API model, no GPU/open weights/corpus).
- No autonomous publish; template ids missing.
- DeepSeek key validity not proven without an owner-approved live call.
- No "better than human designer" claim; no blind study.
- Adobe tutorials not executed; 50-task corpus is a proposal.

- Production migration/implementation is **not** authorized by this plan.

10. Diagram index (validated via Kroki)

Diagram	Purpose	Source
`01-goal-gap`	goals -> code gaps -> phases	`diagrams/01-goal-gap.mmd`
`02-runtime-topology`	VPS services/ports/planes	`02-runtime-topology.mmd`
`03-craft-command-path`	operator -> gateway -> CLI -> verifier -> critic	`03-craft-command-path.mmd`
`04-forge-architecture`	roles, memory, tools, data	`04-forge-architecture.mmd`
`05-learning-state`	episode/skill/rollback state machine	`05-learning-state.mmd`
`06-evidence-ladder`	why each proof rung matters	`06-evidence-ladder.mmd`
`07-decision-tree`	build-vs-buy / scope choices	`07-decision-tree.mmd`
`08-roadmap`	phase gates/time	`08-roadmap.mmd`

11. Open questions only the owner can answer

1. Is the live DeepSeek key valid, and may we spend a minimal completion to prove it? **Preferred: use the zero-spend metadata check first** (`GET /user/balance`, `GET /models`); a chat completion is not a key test.
2. Which Craft app is the first commercial target: VectorCraft or PhotoCraft?
3. Are FilmCraft/DesignCraft/EffectCraft available from a private artifact store, or out of scope?
4. Who is the human approver for print/publish and skill promotion?
5. What are the rights for the first 10/50 golden assets?
6. Is "Harness 2" the Google Harness? project, or an internal second harness? (terminology)
7. Is the desired interface chat-only, or an operator UI with approvals?
8. What is the cost ceiling per artifact/day?

Verdict

The product is buildable now only as a narrow, evidence-gated operator (VectorCraft + PhotoCraft). The model and harness already exist; the blockers are execution wiring, verification, ledger/approval, and scope honesty. Phase 0 must fix the execution path before any "learning" work is meaningful.